

Small quasigroups and the implication semilattice of Schröder's 990 laws

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Abstract

Le Floch [3] determined all implications among Schröder's 990 quasigroup laws: 47 equivalence classes and 114 closed sets. We measure how much of this structure is witnessed by quasigroups of small order. Quasigroups of order at most 4 witness 91.7% of the non-implications between classes, and those of order at most 6 witness 95.2% and realise 94 of the 114 closed sets. Exhaustive search over the models of the remaining laws and closed sets extends the picture exactly: quasigroups of order at most 9 witness 100.0% of the non-implications and realise 114 closed sets, with least witness orders up to 9. A set of 17 quasigroups witnesses every non-implication between classes and, given Le Floch's list of closed sets, no smaller set of quasigroups of any order does. Code and data: github.com/carlok/quadratula.

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MSC 2020: 20N05, 08B05.

1 Introduction

Schröder [6] considered the equations with three variables and two operations on each side, such as $x \circ_1 (y \circ_2 z) = \alpha \circ_3 (\beta \circ_4 \gamma)$, where each $\circ_i$ is independently one of the quasigroup operations $*$, $//$, $\backslash$, either side may instead be bracketed as $(\cdot \circ \cdot) \circ \cdot$, and (α, β, γ) is a permutation of (x, y, z) . Le Floch [3], following the methods of the Equational Theories Project [1], determined all implications among the resulting 990 laws with Prover9 and Mace4 [4]. He states that every non-implication has a finite countermodel, without giving their orders.

We measure, for each k , the part of this structure witnessed by quasigroups of order at most k : non-implications, separation of classes, and closed sets realised as intersections of signatures. Le Floch's result is the ground truth and is not revised.

2 Definitions

A quasigroup of order n is a Latin square on $\{0, \dots, n-1\}$. Following [3], $a // b$ is the c with $c * a = b$, and $a \backslash b$ is the c with $b * c = a$. For a quasigroup Q , $\text{sig}(Q)$ is the set of laws Q satisfies. Q witnesses $A \not\Rightarrow B$ if $A \in \text{sig}(Q)$ and $B \notin \text{sig}(Q)$.

Definition 1. The floor at order k is the set of quasigroups of order at most k up to isomorphism. A non-implication is witnessed at order k if some quasigroup of the floor witnesses it. A closed set V is realised at order k if $V = \bigcap \{\text{sig}(Q) : V \subseteq \text{sig}(Q), |Q| \leq k\}$, with the empty intersection equal to the set of all laws.

Every signature is closed under consequence, including consequences of conjunctions, hence one of the 114 closed sets listed by Le Floch. Equivalent laws have equal signatures, so the partition of laws by signature over a floor is coarser than or equal to the partition into classes.

Lemma 1. *Whether $A \not\Rightarrow B$ is witnessed at order k depends only on the models of A of order at most k . Whether V is realised at order k depends only on the models of V of order at most k , and a closed set realised at order k is realised at every larger order.*

Proof. The first two claims follow from the definitions of witness and realisation (Definition 1). For the third, adding models of V can only shrink the intersection, which always contains V . $\square$

3 Method

Enumeration. Quasigroups of order $n \leq 6$ are enumerated by backtracking over Latin squares, keeping the row-major lexicographically least table of each isomorphism class. Two necessary conditions for leastness prune the search: $T_{00} \leq 1$, and row 0 is least among its conjugates by permutations fixing 0. Each law is checked on the n^3 assignments, stopping at the first failure. Order 6 takes 1.0s to enumerate and 6.1s to evaluate on 11 threads of an Apple M3 Pro.

Verification. (i) The class counts 1, 1, 5, 35, 1,411, 1,130,531 match OEIS [A057991](#) [5], and $\sum n!/|\text{Aut}|$ equals the number of Latin squares ([A002860](#)) at each order. (ii) No floor quasigroup refutes a listed implication (0 cases), and every signature is a listed closed set. (iii) The counts of groups, abelian groups, commutative, semisymmetric, totally symmetric and idempotent quasigroups agree with OEIS ([A000001](#), [A000688](#), [A057992](#), [A076017](#), [A076019](#), diagonal of [A058175](#)): all 36 values match. (iv) The six parastrophes map the law list to itself; the predicted permutation of signatures was checked on every Latin square of order at most 5 (961,513,740 bit comparisons, 0 mismatches). (v) A separate script, with its own law parser and evaluator and reading Le Floch’s files directly, re-checks both covers below and finds a witness of the stated least order for all 94 residual pairs of the Results section. SAT encodings solved with kissat confirm the least orders: Sch-38 and Sch-79 have no model of order 7, no model of Sch-23 of order 7 or 8 violates Sch-5 (which implies Sch-29, Sch-93 and Sch-183), and witnesses exist at orders 8 and 9, re-checked after decoding. Two queries on associativity and commutativity at orders 5 and 6 test the encoding. All 9 queries give the expected answer, each within 20s.

Beyond order 6. By Lemma 1 it suffices to enumerate, at order k , the models of each closed set not realised at order $k - 1$. This also covers witnessing: if $A \not\Rightarrow B$ is not witnessed at order $k - 1$, then B lies in every signature of order at most $k - 1$ containing A , so the closed set generated by A is not realised at order $k - 1$ and its models of order k are enumerated. Separation of classes follows from witnessing. The search fills cells row-major and prunes as soon as a required law fails on defined entries, with the same symmetry breaking and leastness test. With no laws, and with each of seven sample laws, it returns exactly the enumerated isomorphism classes at every order at most 5. As an external check, Sch-5 is equivalent to semisymmetry [3] and implies Sch-23; filtering the Sch-23 runs by Sch-5 gives 41, 595 and 26,620 quasigroups at orders 7, 8 and 9, the values of [A076017](#). The other runs have no such external count.

Cover. Minimum sets of quasigroups witnessing a given set of non-implications are computed as set-cover integer programs with HiGHS [2].

4 Results

Table 1 gives the cumulative counts.

k	class non-impl. witnessed	share	classes separated	closed sets realised
3	1,245	63.6%	0	12
4	1,795	91.7%	30	61
5	1,834	93.7%	36	91
6	1,864	95.2%	42	94
7	1,864	95.2%	42	95
8	1,954	99.8%	45	106
9	1,958	100.0%	47	114

Table 1: Cumulative results for the floor at order k . Denominators: 1,958 non-implications between classes, 47 classes, 114 closed sets. Orders 1 and 2 witness nothing. A class is separated if no other class has the same signatures. Rows 7–9 come from the search justified by Lemma 1 and are exact up to order 9.

Small orders. Counted over laws, the floor at order 4 witnesses 95.3% of the 726,207 non-implications, and the floor at order 6 96.7%. The unique quasigroup of order 2 satisfies all laws. $\mathbb{Z}_3$ satisfies 150 laws and alone witnesses 17.4% of the law non-implications.

Residual. At order 6, 94 class non-implications are unwitnessed, all with hypothesis Sch-23, Sch-38 or Sch-79. Their least witnesses have order 8 (Sch-38), 8 (Sch-79) and 9 (Sch-23); Mace4 agrees for 94 of the 94 pairs. Small quasigroups miss these pairs because the hypotheses have few models; for $n = 1, 2, \dots$ the model counts are Sch-7: 1, 1, 0, 1, 0, 0, 0, 1, 0; Sch-38: 1, 1, 0, 1, 0, 0, 0, 3; Sch-5: 1, 1, 2, 3, 4, 9, 41, 595, 26620; Sch-23: 1, 1, 2, 3, 4, 9, 41, 595, 26623. Sch-7 implies Sch-38, so equal counts mean equal sets of models: up to order 7 no quasigroup separates them.

Closed sets. Least realising order of the 20 closed sets not realised at order 6: order 7: 1; order 8: 11; order 9: 8. Not realised at order at most 9: none.

Covers. The 1,864 class non-implications witnessed at order at most 6 are witnessed by 17 quasigroups (4 of order 3, 4 of order 4, 6 of order 5, 3 of order 6), optimal among quasigroups of order at most 6; greedy uses 20 and the LP bound is 17. For all 1,958 class non-implications, the optimum among known signatures is 17. An ILP over all 114 closed sets as hypothetical signatures gives the lower bound 17, valid for quasigroups of any order provided Le Floch’s list of closed sets is complete.

Symmetry. Under the parastrophes the 990 laws form 183 orbits, the classes 20, the closed sets 46 and the ordered class pairs 550.

Model counts. Counts of models by order for all classes and closed sets are in the repository. Of the 10 distinct class count sequences for orders 1–6, searching OEIS for the terms together with “quasigroup”, “latin” or “loop” finds A057992, A076017, A076018, A076019; for 7 of them these searches return nothing.

5 Limitations

The ground truth is Le Floch’s computation and is not re-proved here. In particular, the lower bound for covers assumes that his 114 closed sets are all the subsets of laws closed under

consequence. Least witness orders beyond 6 are confirmed by SAT; the realisation of closed sets at orders 7–9 rests on the completeness of our search, checked as described. Mace4 is used only for comparison. An OEIS match of six terms of a model-count sequence is not a proof. No result is formalised in Lean.

Data and code

<https://github.com/carlok/quadratula> (Apache-2.0). Le Floch’s ancillary files are redistributed under CC BY 4.0.

Use of AI

Code, analysis and text were produced with Claude (Anthropic) under the author’s direction. All numbers are generated by scripts in the repository.

References

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